# Adversarial Reinforcement Learning for Robust Web Cache Replacement

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***Abstract:*** Cache replacement decides which item to evict when the cache is full, directly determining hit rate in web systems. Classical policies like LRU and LFU use fixed rules; recent ML agents like LeCaR and GL-Cache learn from traffic traces and do better but only on patterns they have seen. We propose RARL-Cache, which co-trains two DQN agents: a protagonist that learns the eviction policy, and an adversary that acts as a training partner by continuously generating request sequences the protagonist handles poorly. This forces the protagonist to generalise beyond any fixed trace. The adversary is discarded at deployment. We evaluate on Wikipedia CDN traces and measure whether RARL-Cache outperforms plain DQN and classical baselines when traffic patterns shift.

***Key Directions and Expected Outcomes***

**Week 1: Dataset Exploration & Classical Baselines**

- Load Wikipedia CDN traces or generate Zipfian synthetic traces
- Implement LRU and LFU baselines
- Establish performance benchmarks
- **Expected Outcome:** Hit rate comparison plots showing LRU vs LFU performance across different workloads as baseline metrics

**Week 2: Single-Agent DQN Training**

- Build DQN agent trained on fixed trace
- Measure performance on training vs. unseen data
- Identify generalization gap
- **Expected Outcome:** Demonstrate single-agent overfitting problem, DQN matches or beats LRU hit rate on the same Wikipedia traces

**Week 3: Adversarial Co-Training**

- Add a second DQN agent that controls which URLs get requested, trained to minimize the protagonist's hit rate
- Co-train both agents with alternating updates and progressive difficulty
- **Expected Outcome:** Training curves showing protagonist hit rate drop as adversary improves, then recovers as protagonist adapts (visualizing the core RARL dynamic)

**Week 4: Comprehensive Evaluation**

- Compare against: LRU, LFU, DQN, LeCaR, GL-Cache
- Evaluate robustness under distribution shifts
- Conduct ablation studies (with/ without adversary)
- **Expected Outcome:** RARL-trained protagonist holds a higher hit rate than any other approaches when traffic patterns shift outside training distribution.

***References:***


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